Advanced Educational Activities Pvt. Ltd.

(Key and Solutions for AIMCAT2405)

Key

SECTION – I

1. A	7. B	13. C	19. B
2. B	8. A	14. A	20. 1324
3. C	9. B	15. B	21. B
4. B	10. A	16. B	22. D
5. B	11. B	17. A	23. B
6. D	12. D	18. 2134	24. 3241

SECTION – II

1. 7	6. C	11. 138	16. C
2. A	7. A	12. 368	17. A
3. 8	8. A	13. D	18. 9
4. D	9. 3	14. C	19. A
5. D	10. B	15. A	20. C

SECTION – III

1. B	7. B	13. A	19. B
2. C	8. 10	14. D	20. 107
3. D	9. 437	15. A	21. 12
4. C	10. B	16. D	22. 150
5. D	11. 64	17. C	
6. 10	12. B	18. C	

Solutions

SECTION – I

Solutions for questions 1 to 4:

Number of words and Explanatory notes for RC:

Number of words: 539

1. Option A: If this is true, then it invalidates the argument against heuristics and snap judgments. The term bounded rationality used by Herbert Simon refers to heuristics and shortcut decisions that people take. If they are as efficient as the ones produced by the optimisation process recommended by decision analysts, then the argument that heuristics adversely affect our decision-making process is invalidated. Option A is the answer.
Option B: While this explains why decision makers go for heuristics-based decisions and judgments, it doesn't really explain that such decisions are rational. They save time, yes, compared to the optimisation process in probability-based decisions and rational analysis. Hence, this option doesn't really invalidate the argument against heuristics. Option B is not the answer.
Option C: This option validates the argument rather than invalidate it. It shows that cognitive biases do affect decisions and it is possible to avoid them to make more rational decisions. This goes against the rationality of heuristics-based judgments. Hence, Option C is not the answer.
Option D: While this option is positive about heuristics and heuristics-based decisions, it doesn't necessarily confirm that heuristics lead to rational decisions. The option just says that heuristics helps avoid over-analysis, which could be true even if heuristics lead to irrational choices. Option D is not the answer. Choice (A)

2. Option A: This is the finding of the study conducted by Gigerenzer. The study by Gigerenzer and co-authors led to this particular conclusion (that goes against what Kahneman and Tversky proposed) that sometimes cognitive illusions may not even exist if participants are asked questions differently. Option A is not the answer.
Option B: Gigerenzer and other co-authors of a study came up with the finding that at least some of the cognitive illusions do go away when questions are framed differently to individuals.

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Consider this: 'He was, first, dubious of some of the results. By tweaking the framing of a question, it is sometimes possible to make apparent cognitive illusions go away.' So, Gigerenzer's study weakens the argument that cognitive illusions are always playing a role in decision-making. Because the study tries to prove that at least sometimes, cognitive illusions can go away. Option B is the answer.
Option C: This is Herbert Simon's argument about 'bounded rationality' and it doesn't concern the framing of a question that Gigerenzer and co-authors of the study felt would eliminate cognitive illusions. So, Option C is not the answer.
Option D: This is the opposite of Gigerenzer's argument which is that heuristics and other methods humans use shouldn't be dismissed as inferior. However, this argument is not really affected by the example of doctors and patients which is more about how cognitive biases may actually manifest because of the framing of a question, and disappear when questions asked to decision makers/study participants are framed differently. That example doesn't say anything about whether heuristics are good or bad. Option D is not the answer. Choice (B)

3. Option A: The former (Simon) didn't really argue that people didn't depart from the rational model. He argues that the rational model wasn't actually the best. In other words, Simon disputed the idea of the rational model. That's why he advocated the term bounded rationality. Option A is not the answer.
Option B: Herbert Simon didn't really argue that there are multiple rational models, neither did Kahneman and Tversky. So, the point that 'not all' rational models provide the best course of action is not accurate. It indicates that some rational models may provide the best course of action. Simon disputed that 'the rational model' is the best. Similarly, Kahneman and Tversky didn't argue that rational models are not followed. They only spoke about individuals departing from the 'rational model.' Option B is not the answer.
Option C: This is apt as understood from 'Simon's "bounded rationality" is often depicted as a precursor to the work of Kahneman and Tversky, but it was different in intent. Whereas they showed how people departed from the rational model for making decisions, Simon disputed that the "rational" model was actually best.' So, Simon's argument was that the rational model was not necessarily the best, and, therefore, his work was not really an accurate precursor to Kahneman and Tversky's work, which said that people depart from the rational model because of their heuristics and biases. They were different in what they were trying to prove. Option C is the answer.
Option D: Simon didn't really advocate that rational models work well with heuristics. In fact, we are not even sure such a thing is possible, since the rational model by its very nature is something that is free of heuristics. Also, the discussion of mixing rationality with heuristics 'consciously' has not been brought up in the passage. Option D is not the answer. Choice (C)

4. Option A: This option, if true, shows that gut feeling is usually right/accurate. That shows that going by heuristic biases doesn't always make an individual a cognitive cripple. This option doesn't weaken the fatalistic view. Option A is not the answer.
Option B: If true, this option explains why we take decisions based on heuristics. But it doesn't tell us whether such decisions are good or bad. Why they are taken doesn't tell us anything about the future outcome of such decisions. So, this option neither goes in favour nor against the idea that heuristics are bad for us. Hence, Option B is the answer.
Option C: This option shows that fast decisions based on gut, instinct and heuristics can be taken under pressure, and such decisions could have good outcomes. That weakens the argument of the fatalistic view that heuristics turn individuals into cognitive cripples. Option C is not the answer.
Option D: If true, this statement favours heuristics by showing that they actually save time and resources even if we lose a little efficiency along the way. So, it weakens the fatalistic view that heuristics lead to cognitive bias and make us take bad decisions. Option D is not the answer. Choice (B)

Solutions for questions 5 to 8:

Number of words and Explanatory notes for RC:

Number of words: 530

5. Option A: The hemispherical foil isn't coated with aluminium oxide to function like a curved retina. This can be understood from 'Then they treated the metal with an electrochemical process that converted it into an insulator called aluminium oxide.' The foil itself is converted to aluminium oxide. Hence, Option A is not the answer.
Option B: This can be understood from 'The research built on the fact that ***perovskite, a conductive and light-sensitive material used in solar cells, can be used to create extremely thin nanowires*** several thousandths of a millimeter in length. These wires mimic the structure of the eye's long, thin photoreceptor cells, says study co-author Zhiyong Fan.' Perovskite nanowires produce/generate charge as a response to patterns of lights because they are light-sensitive. Hence, Option B is the answer.
Option C: The passage doesn't mention that perovskite nanowires can generate charges only when they are immersed in an ionic salt. They are photoreceptors, so they can generate charge based on patterns of light. The ionic salt only helps the charge move to facilitate the electric exchange. Option C is not the answer.
Option D: The nanoscale pores in aluminium foil do not 'assist' in the generation of perovskite nanowires. This can be understood from 'that had convenient densely clustered holes in which they could "grow" perovskite nanowires...' The researchers use the holes to 'grow' the nanowires in them. Hence, Option D is not the answer. Choice (B)

6. Consider this: 'Inspired by the specialized liquid in a real eye, the team filled its biomimetic version with an ***ionic liquid, a type of liquid salt in which charged particles can move.*** "One very important component inside is in the cavity we filled [with] ionic liquids," Fan says. "***Once these nanowires generate charges, the charge will be exchanged with some ions.***" This electrical exchange allows the perovskite nanowires to perform the electrochemical function of detecting light and sending that signal to external image-processing electronics.'
Option A: The charges are not created by the perovskite nanowires. They are merely carried by the ionic liquid. The charge is actually created by the nanowires which act like photoreceptors, generating charge from light patterns. Option A is not the answer.
Option B: The ionic liquid doesn't really generate signals to external image-processing electronics. The signals are in effect generated by the nanowires, and these charges are exchanged with some ions so the electric exchange can be transferred as a signal to the external image-processing electronics. Option B is not the answer.
Option C: While this is indeed true that the ionic liquid fills the biomimetic version of the eye, the function of the liquid is not just to fill the eye. It is to transfer the charges generated by the nanowires to the image-processing electronics. Hence, Option C is not the answer.
Option D: The function of the ionic liquid is to facilitate an electrical exchange so that the perovskite nanowires perform the electrochemical function of detecting light and sending that signal to external image-processing electronics. They facilitate such an exchange because once nanowires generate charges, the charge will be exchanged with some ions and that's how the charge travels. The charge itself is generated by the nanowires based on light patterns. Option D is the answer. Choice (D)

7. Option A: Consider this: 'In some ways, the artificial eye improved on natural vision: it could pick up a greater range of wavelengths and lacked a blind spot.' From this, we can understand that the artificial eye is an improvement on natural vision because it lacks a blind spot. So, it can be inferred that there is a blind spot or two in the natural vision. Option A is not the answer.
Option B: The comparison of contrast and clear edges was between the biomimetic eye and an artificial eye with a flat image sensor. This is not the comparison between artificial eye and natural eye. This option cannot be inferred. Hence, Option B is the answer.
Option C: This is true and can be understood from 'In some ways, the artificial eye improved on natural vision: it could pick up a greater range of wavelengths and lacked a blind spot.' The artificial eye can pick a greater range of wavelengths compared to natural vision. Hence, Option C is not the answer.
Option D: This can be understood from 'When the team tested the artificial eye, it managed to process patterns of light in as little as 19 milliseconds—half the time required by a human eye.' Processing speed of the artificial eye is higher compared to the human eye. Option D is not the answer. Choice (B)

8. The retina is curved. Similarly, the sensors in the artificial eye have to be curved for a sharp image, as understood from 'When you try to image something, the image that forms after the lens is actually curved...' and 'If you have a flat sensor, then the image cannot be focused very sharply... the retina is curved, but electronic light sensors are rigid and flat.'
Option A: This is true as flat, rigid sensors cannot replicate the functioning of a curved retina. The flat sensors do not produce a clear image. Hence, Option A is the answer.
Option B: While the sensors have to be curved and not flat is because of the clarity of the image. Forming an artificial hemispherical retina was the first concern. How to grow nanowires in it is the second concern. The first concern is what the question is about. Hence, Option B is not the answer.
Option C: While this is true, it explains how a curved retina helps form the right image. This option, however, doesn't explain why the flat sensors are a problem. Hence, Option C is not the answer.
Option D: The sensors need to be hemispherical in order to form a focused image, as is the case in a curved retina. How the perovskite nanowires are grown in a hemispherical artificial retina is the next case, and is not the concern of the question. Option D is not the answer. Choice (A)

Solutions for questions 9 to 12:

Number of words and Explanatory notes for RC:

Number of words: 526

9. Option A: While the relationship between sleep, dreaming and creativity has been the subject of discussion/speculation/conjecture, the para doesn't say that it was actually studied during this entire time period. Hence, Option A is not the answer.
Option B: This is the purpose of the first para. The para establishes the idea that for hundreds of years the relation between dreaming and creativity was the subject of a conjecture, was speculated upon. The example of Edison proves that this was an area of interest for quite a while. The next para offers scientific evidence. The first para basically builds up to the idea that there was conjecture/discussion of the idea. So, the conjecture was not new. Hence, Option B is the answer.

Option C: The first para doesn't suggest that Edison's method was scientific. The scientific studies were mentioned only in the second para. So, the purpose of the first para was not to give scientific corroboration of any kind to the connection between dreaming and creativity. Option C is not the answer.
Option D: The para was not about Edison in particular (neither is the rest of the passage). Also, the para doesn't really say that Edison was the first to make the connection between creativity and dreaming. The para merely gives the example of how Edison mined his dreams for creative ideas. Option D is not the answer. Choice (B)

10. Option A: The success of the study depends on identifying the usefulness of the solutions provided by the participants after the incubation in their sleep. If the solutions cannot be tangibly compared, then it is hard to correlate frequency and complexity of recall with higher creativity scores, since it is understood that creativity scores will be high when the solution is more useful. Hence, this answer needs to be evaluated to determine the success of the study. Option A is the answer.
Option B: The frequency of dream recall shouldn't be consistent amongst participants. In fact, they vary from one participant to another, which is why a correlation to usefulness of the solution can be discovered. More frequency and better detailing should correlate to better solutions. Option B is not the answer.
Option C: This has been mentioned to be standardized as understood from '...have been correlated with higher scores on creativity evaluations...' Unless the criteria are standardized, there cannot be higher and lower scores. It must also be understood that irrespective of what the criteria are, they are consistent for all participants. So, this answer need not be provided, it is already understood. Option C is not the answer.
Option D: The para itself provides the answer that REM sleep begins about 70 minutes after a person loses consciousness. So, this question doesn't determine the success of the study which was conducted to prove the connection between creativity and REM sleep. More than the 'when' of the REM sleep, the observation has to be about what the quality of the dream was and if it can be assessed. Option D is not the answer. Choice (A)

11. Option A: The study replicated Thomas Edison's method. However, ***the study doesn't establish a connection between dreams and creativity.*** It establishes a connection between N1 sleep stage and creativity. Hence, Option A is not the answer.
Option B: This is the conclusion that the study supports as understood from 'She found that those who woke from the earliest stage of sleep were far more likely to solve the problem.' N1 sleep onset does 'influence the creativity' of individuals. The study shows that individuals waking up from N1 sleep state were able to tackle the problems better/more innovatively. N1 state of sleep is where creativity happens and, hence, waking up from it, generally leaves individuals remembering greater details and coming up with innovative solutions. Hence, Option B is the answer.
Option C: N1 is the earliest stage as understood from 'N1 is a hybrid, or "semi-lucid" state of mind, says French neuroscientist Celia Lacaux, when individuals are just beginning to detach from the waking environment.' Waking up from this stage was correlated to most innovativeness. So, by extension and based on '***This shadowy frontier between waking and dreaming... may be the source*** of many of humanity's most novel ideas,' this state of mind could be called the breeding ground for innovativeness. REM sleep relates to the lucid state. So, the lucid state of mind is not the breeding ground for innovations as understood from the passage, it is the semi-lucid state. Hence, Option C is not the answer.
Option D: There is a correlation between creativity and waking up ***specifically*** from the earliest stage of sleep which is N1 sleep onset. The correlation between creativity and ***waking up from any other stage of sleep*** has not been established from this study and is, hence, a faulty generalisation. Option D is not the answer. Choice (B)

12. We are looking for an option that doesn't explain why the author thinks there is a definite connection between creativity and hypnagogia. 'Intuitive sense' means that it is logical why one would be creative in hypnagogia.
Option A: The last para defines creativity as 'One major theory of creativity posits that it results when our minds make connections between distantly related concepts stored in our memories.' Since, this is very similar to what happens during hypnagogia, the author believes it is logical to conclude that the two are connected. Hence, this option explains why the author felt it makes intuitive sense to connect creativity and hypnagogia. Option A is not the answer.
Option B: This option provides further explanation as to why the author feels there is a logical connection between why creativity works better in hypnagogia and why it makes sense that in that state of sleep, innovativeness is fuelled. That's because creativity is making unusual connections and that tends to happen in hypnagogia. Option B is not the answer.
Option C: Consider this: 'this mashing together of old and new is a process that helps to minimize redundancy and complexity in our memory system, and ***prepares us to navigate a fuller range of possible scenarios in our waking lives***...' So, the mashing up of old and new makes us more perceptual of distinct possibilities which could be unique and creative, and not lost in the chaos of redundant situations. So, this option indeed explains why the author thinks the connection between hypnagogia and creativity makes sense. Option C is not the answer.

Option D: The dreamer ***doesn't really control the external environment.*** The dreamer is closer to the surface of sleep, ***to conscious control and to the external perceptual environment.*** So, this option misrepresents information. The dreamer in hypnagogia is close to the surface of sleep while also to the external environment leading to the mixing of memories and experiences. The dreamer doesn't control anything, per se. Directing of dreams may happen in case of lucid dreams in REM sleep as mentioned earlier in the passage, but N1 is a semi-lucid state. Hence, Option D is the answer. Choice (D)

Solutions for questions 13 to 16:

Number of words and Explanatory notes for RC:

Number of words: 521

13. Option A: Political and legal constitutionalism is about which institution has the higher authority. While it is true that we can either have legal constitutionalism or political, and not both, or have an overlap, because of the way the terms are defined, the option doesn't explain what the author meant by organ sovereignty, which is that only one of the institutions can be sovereign. Hence, Option A is not the answer.
Option B: The concept of organ sovereignty is not about balancing power. It is about who will be the ultimate/final authority and, hence, by extension truly autonomous. While one takes that seat the other can't. So, the term is not about balancing power, but about which institution has the ultimate sovereignty. Option B is not the answer.
Option C: This explains the concept of organ sovereignty in the sense that only one institution can be sovereign, whether legislative or judicial. The other will at some point have to revert to this institution for guidance or the final word. In legal constitutionalism the judicial institution will have the final word, and in political constitutionalism, the legislative institution will have the final word. Both can't really share the power balance. Option C is the answer.
Option D: The concept of organ sovereignty is about how one institution – judicial or legislative – will be sovereign and hence cannot be dictated terms to by any other institution. It is not a comparison of powers between two institutions, as to which institution is more powerful. The focus here is on which institution will be the sovereign one and which institution will have to look up to the other for the final word. Option D is not the answer. Choice (C)

14. The legalist claim is that the constitution is given authoritative meaning by judicial interpretation in the way in which the court interprets non-constitutional laws. So, an option that undermines the judicial interpretation and places the constitution above such an interpretation will weaken this claim.
Option A: This option weakens the legalist claim because it says that the constitution's authority is final and definite, and the judicial interpretation doesn't matter. In other words, the constitution is not subject to judicial interpretation. That weakens the legalist claim as it contradicts the claim of the authority of judicial interpretations. Option A is the answer.
Option B: Even if this were true, it doesn't negate the legalist claim, that the constitutional authority is itself provided by the judicial interpretation (how the judicial body interprets the laws defines the constitution according to the legalist claim). Option B is not the answer.
Option C: Even if such a challenge is possible, it will add to the argument of the political constitutionalists but will not weaken the claim of the legalists. This option doesn't say that the legal constitutionalists' opinion (that their interpretation determines what the constitutional viewpoint is) is wrong. It only says such a viewpoint may be challenged. Hence, Option C is not the answer.
Option D: Even when the courts can do that, it still stands that their interpretation determines the constitution's authority. So, this option does not take away anything from the legalist view, and, actually, propagates it by saying that the constitution's interpretation depends on how the judicial interpretation stands. Hence, Option D is not the answer. Choice (A)

15. Option A: The author hasn't said here that constitutional order necessarily leans towards the judicial institution. The author says that the state acts through its '***designated constitutional organ that can speak authoritatively.***' That could be legislative or judicial. Hence, Option A is not the answer.
Option B: This is apt since the author talks about the state deciding through its designated constitutional organ. That organ could be judicial in case of legal constitutionalism. It could be the legislative institution in the case of political constitutionalism. Either way, it is the constitution that will be the guiding light. Hence, the term constitutional order has been used. Living in the constitutional order is basically living according to the constitutional law as interpreted by the designated constitutional organ. Option B is the answer.
Option C: The author doesn't say that there are multiple constitutional organs that interpret the constitution. There is one institution that will have the authority, the sovereignty needed to implement the law of the constitution. Also, it is the state that acts through the authoritative, designated constitutional organ. Also, the laws are not necessarily framed by the organs. It is more interpretation of the constitutional and non-constitutional laws that is the imperative for the state. Hence, the author won't approve of this statement. Option C is not the answer.
Option D: The author doesn't say that living in a constitutional order means a balance between judicial and political processes. It is more the state acting through its designated constitutional organ that determines the right interpretation of the law provided by the constitution. Hence, Option D is not the answer. Choice (B)

16. The drawback has been pointed out by the author as: 'Nevertheless, owing to their shared assumption of organ sovereignty, ***both political and legal constitutionalism fall short in accounting for the dynamics of constitutional ordering*** and thus render a limited explanation of how law and politics interact constitutionally.'
Option A: While this is true, that only one of the two constitutional organs can be sovereign (and the other has to concede to it as the authority), this is not the disadvantage/drawback pointed out by the author. The drawback is that both forms of constitutionalism fail to account for the dynamics of constitutional ordering. Option A is not the answer.
Option B: This explains the drawback. The author pointed out that both forms of constitutionalism fail to render a full explanation of how law and politics interact constitutionally. In other words, neither constitutionalism fully explains the dynamism of interaction between law and politics as laid out in the constitution. Option B is the answer.
Option C: While constitutional governance is indeed dependent on a dynamic socio-political process, the author doesn't say that the shared assumption hinders such a process. The author in fact says that 'Put simply, the assumed organ sovereignty is only a function of a continual dynamic socio-political process that has resulted in constitutional governance.' That means constitutional governance goes on regardless. Hence, Option C is not the answer.
Option D: The shared assumption means that neither institution (political or judicial) attains the seat of sovereignty. However, the drawback of the shared assumption is not that it ***particularly stops these institutions from designated authorities.*** The debate means that there is no designated authority per se, as an obvious consequence. The drawback of the debate is what the question focuses on here, and that drawback is the insufficient accounting of the dynamic process that fuels constitutional governance. Option D is not the answer. Choice (B)

Solution for question 17:

17. Option A: This option sums up the main idea of the para, that statistical illiteracy influences the emotional climate in a society negatively because such a climate (as a sum total of the hopes and anxieties) can be manipulated by external forces. Option A is the answer.
Option B: Collective statistical illiteracy doesn't make the emotional climate resistant to manipulation by external forces. It is literacy that is the precondition. This option mentions that it is illiteracy that makes things better. Option B is not the answer.
Option C: Collective statistical literacy prevents hopes and anxieties from getting manipulated. It doesn't protect the 'society' from anxiety per se. Also, this option doesn't mention the emotional climate of a democracy that statistical literacy influences. Option C is not the answer.
Option D: Collective statistical literacy is a ***precondition*** for an educated citizenship in a technological democracy. It doesn't 'create' a technological democracy. Hence, Option D is not the answer. Choice (A)

Solution for question 18:

18. Sentence 1 is an independent sentence that highlights how lawyers 'read closely' because of professional responsibility.
Sentence 2 is also an independent sentence that highlights the ingredients of the 'lawyer's craft' – a willingness to immerse herself within the fine print.
Sentence 3 refers to a 'skill' in the previous sentence, a skill that helps lawyers take responsibility.
Sentence 4 describes the fit between fledgling startups and lawyers.
It should be noted that 'such a skill' in 3 has to be preceded by the discussion about a skill. It can also be noted that both 1 and 2 talk about the same skill – the ability to read fine print/read closely. But it is sentence 2 that defines the term 'fine print' and 1 elaborates the same. That means 21 is the order and not 12. 3 follows the 21 block because it refers to this particular skill of reading closely. 4 comes in the end because it talks about where lawyers will be needed to use such a skill – more like an application of this ability. So, the order is 2134. Ans: (2134)

Solution for question 19:

19. The paragraph has two main ideas: visuospatial and verbal storage mechanisms are not associated with each other. They are distinct from each other. Secondly, there is information about the verbal storage mechanisms but not much about spatial memory mechanisms.
Option A: Evidence shows that these two memory mechanisms are distinct. It doesn't argue that they are 'completely different.' Also, this option leaves out the second important idea given above. Option A is not the answer.
Option B: This option provides both the ideas of the para, that spatial and verbal storage mechanisms are delineated, different/dissociated. There is substantial evidence supporting this quite strongly. Also, plenty of work has gone into understanding verbal storage mechanisms, but spatial memory mechanisms are not well-understood (description is elusive). Option B is the answer.
Option C: The para doesn't say that spatial memory is based in processes 'that are more difficult to understand'. It simply says that a description about these processes has been elusive. That could also be because not much research has gone into spatial memory mechanisms, or at least as much as the research in verbal storage. Option C is not the answer.
Option D: Firstly, it is incorrect that neuropsychological evidence has 'confirmed' the separability. There is substantial evidence to suggest that the two processes (spatial and verbal storage mechanisms) are dissociated. Also, lack of information is not a problem for verbal storage. But the option says there is a lack of information about 'their' component processes, meaning both types of mechanisms. Hence, Option D is not the answer. Choice (B)

Solution for question 20:

20. Sentence 1 is an independent sentence that talks about the cognitive factors involved in crowd evacuation behaviours.
Sentence 2 provides an explanation of such a crowd evacuation behaviour. So, it can be understood that 2 is downstream to 1.
Sentence 3 mentions how objective data are lacking, and, hence, the effects of 'these factors' hasn't been clarified. It can be understood that 'these factors' refers to the cognitive factors in 1. So, 13 is a block.
Sentence 4 refers to the video clip and discusses the analysis that follows and what researchers discovered from it. So, 4 has to follow 2. Since, 13 is about the theory, and 24 is experimental evidence, the right order will be 1324. Ans: (1324)

Solution for question 21:

21. Option A: It cannot be the first blank because moths' drabness in normal light is introduced afterwards, followed by a 'But' in the next sentence. That sentence has the same tone as the bold sentence. So, 1 is not the right place.
Option B: This is the right place for the bold sentence, as it comes after the tone change in the previous sentence that talks about how moths look different under infrared. The blank is followed by what happens in infrared. So, the second blank is the right place for the bold sentence.
Option C: This can't be the right place for the bold sentence because the bold sentence talks about how the moths were photographed, while the sentence before the third blank is already talking about the observations. So, observations will only come after the photography. Option C is not the answer.
Option D: The sentence before the fourth blank talks about 'new data' which was obtained from the photography. So, the bold sentence will come much before the fourth blank. Option D is not the answer. Choice (B)

Solution for question 22:

22. Option A: At this point, the faults have been introduced, but we are still talking about friction in the next sentence and what happens along the fault. So, seismic waves rippling is much further downstream and doesn't fit here. Option 1 is not the answer.
Option B: We can understand that there is no statement in Blank 2 because the two sentences on either side are a continuing idea: friction causes the plates to get stuck and tension builds up till the fault snaps. Option 2 is not the answer.
Option C: The sentence following the third blank talks about the lurching movement of the plates. Unless this happens we cannot get to the seismic ripple, which is an effect of this movement. Hence, Option 3 is not the answer.
Option D: This is correct position for the given sentence. The preceding sentence talks about an earthquake being unleashed, the cause of the ripple. The subsequent sentence talks about 'they' reaching the earth, they being seismic ripples. This is consistent with the pronoun 'they' as well. Option 4 is the answer. Choice (D)

Solution for question 23:

23. Option A: The para doesn't say that the ease of doing business is 'misunderstood' by American companies. The misconception is about India being politically homogenous. This option also blames the American companies for misunderstanding how investing works. The para is more like a suggestion to American companies as to how to go about investing in India. Option A is not the answer.
Option B: This option has both the ideas in the para: that there is variation in laws across states in India, and that American companies should do their due diligence before setting up shop as ease of doing business varies. The subsidiary ideas about the misconception of India's political homogeneity or the similarity with European ideas are missing, but the option is concise and maintains the essence and tone of the para. Option B is the answer.
Option C: This option only focuses on the similarity between India and the European Union regarding how different states have different laws. It doesn't talk about investing for American companies and the author's suggestion for such companies to do the due diligence. These are more important ideas than the similarity with the European Union. Option C is not the answer.
Option D: The para never says that the misconception about India's homogeneity hides the variation in state laws. Such a link has never been established. The author simply points out that American companies investing here should be cognizant of the variation in state laws. Option D is not the answer. Choice (B)

Solution for question 24:

24. Sentence 1 points to a previous sentence and reports a 'crucial aspect of power relations between coloniser and colonised. This sentence has to follow a sentence that represents something to do with colonisation.

Sentence 2 starts with a positive connector and reports that the English themselves were Anglophones for most of their history.
Sentence 3 is an independent sentence about how English is a hybrid language.
Sentence 4 talks about a different chronological event starting with 'later' when English became relational to all other languages.
The 'later' in Sentence 4 will make sense only if it is contrasted with an 'earlier' which we find in Sentence 2 in the form of 'for most of their history'. The 24 block explains the transition of English from being anglophone to becoming relational to all other languages. This block is an extension of Sentence 3 that introduces the idea of English being a hybrid language. Similarly, 1 has to follow 4 to justify the relationship between coloniser and colonised as foreshadowed in 4 which talks about the British Empire and introduces the idea of what the Indians, Africans and Native Americans spoke. So, the right answer is 3241. Ans: (3241)

Difficulty level wise summary - Section I	
Level of Difficulty	Questions
Very Easy	–
Easy	6, 8, 21
Medium	3, 5, 7, 9, 10, 11, 12, 19, 20, 22, 23
Difficult	1, 2, 4, 17, 18
Very Difficult	13, 14, 15, 16, 24

SECTION – II

Solutions for questions 1 to 5:

Given the number of goals scored in the three matches by the team form either three consecutive odd integers or three consecutive even integers.
Since Felons scored 24 goals, it must have scored 6, 8 and 10 goals in the matches.
Since Lemons scored 15 goals, it must have scored 3, 5 and 7 goals in the three matches.
Riots must have scored 7, 9 and 11 goals.
Dealers must have scored 2, 4 and 6 goals.
Since each team won at least one match, Dealers must also have won at least one match. The only way Dealers can win a match is if they score 4 or 6 goals and the opposing team scored less than 6 goals. Only Lemons scored less than 6 goals in any match. Hence, Dealers must have scored 4 or 6 goals against Lemons and Lemons must have scored 3 or 5 goals.
Lemons also has to win a match and it must have won a match against either Felons or Riots. However, Lemons could not have won a match against Riots because Lemons scored a maximum of 7 goals in a match and Riots scored a minimum of 7 goals in a match. Hence, Lemons must have won against Felons. This is possible only if Lemons scored 7 goals and Felons scored 6 goals in that match. Since Felons scored 6 goals against Lemons, the other two teams must have scored 23 – 6 = 17 goals against Lemons. Dealers could have scored 4 or 6 goals against Lemons. If Dealers scored 4 goals, Riots must have scored 13 goals which is not possible. Hence, Dealers scored 6 goals against Lemons and Riots scored 11 goals against Lemons. Against these two teams, Lemons must have scored 3 and 5 goals in any order and would have lost both the matches.
Since Lemons scored 7 goals against Felons, Riots and Dealers must together have scored 9 goals. This is possible only if Riots scored 7 goals and Dealers scored 2 goals. Felons would have won both the matches since they scored 8 and 10 goals in the two matches in any order.
Since Dealers scored 2 goals against Felons and 6 goals against Lemons, they must have scored 4 goals against Riots.
Since Riots scored 7 goals against Felons and 11 goals against Lemons, Riots must have scored 9 goals against Dealers.
The following table provides the goals scored by each team in each match that they played in the tournament:

		Goals Scored Against			
		Felons	Lemons	Riots	Dealers
Goals Scored By	Felons	X	6	10/8	8/10
	Lemons	7	X	3/5	5/3
	Riots	7	11	X	9
	Dealers	2	6	4	X

1. Riots scored 7 goals against Felons. Ans: (7)
2. Lemons won exactly 1 match. Choice (A)
3. If Felons scored 10 goals in the match against Riots, Lemons must have scored 3 goals against Riots (for GA of Riots to be 17). In the match against Lemons and Riots, the difference in goals is 8.
If Felons scored 10 goals in the match against Dealers, then in this match, the difference in goals is 8.
Hence, in both the cases, the maximum difference in goals in a match is 8. Ans: (8)
4. The number of goals scored in the match between Riots and Lemons can be 14 or 16. Choice (D)
5. In any of the above cases, Felons scored at least one goal more than Riots. Choice (D)

Solutions for questions 6 to 10:

From (iv), the first person to enter the room could not be Bobby or Kartik or Farhan or Gaurav. Hence, the first person has to be Shiv.

From (iii), the Servant cannot be Bobby (since he entered at 10:40 PM) or Gaurav (since he was the last person) or Farhan. Hence, the servant has to be Kartik. Therefore, Kartik entered the room at 10:30 PM and exited at 10:35 PM.

From (v), the person who exited the bedroom at 10:48 PM cannot be Bobby (since Bobby cannot be the secretary). It also cannot be Gaurav because Gaurav entered the room after 10:50 PM. Hence, it has to be Farhan. From (i), Farhan stayed in the bedroom for five minutes. Hence, Farhan would be the secretary and must have entered the bedroom at 10:43 PM. Therefore, Bobby must be the Butler.

The following table presents the profession and the time at which each person entered and exited the bedroom:

Order	Person	Profession	Time In	Time Out
1	Shiv	Carpenter	10:25	10:28
2	Kartik	Servant	10:30	10:35
3	Bobby	Butler	10:40	
4	Farhan	Secretary	10:43	10:48
5	Gaurav	Driver	After 10:50	

6. The police would have arrested Farhan. Choice (C)
7. Kartik, the servant, was in the bedroom at 10:33 PM. Choice (A)
8. Bobby would have been in the bedroom at 10:42 PM. Choice (A)
9. Three people entered the bedroom before Farhan. Ans: (3)
10. Bobby is the butler. Choice (B)

Solutions for questions 11 to 15:

On Day 1, he sold 142 pens. The cost of the first 54 pens (which belonged to Carton 1) is ₹405. The cost of the next 88 pens (from Carton 2) = $\frac{1000}{125} \times 88 = 704$
On Day 2, he sold 124 pens, of which 34 pens are from Carton 2, and 88 pens are from Carton 3 and 2 pens are from Carton 4.
The total cost of the first 34 pens is ₹272, the cost of the next 88 pens is ₹594 and the cost of the next 2 pens is
$\frac{481}{74} \times 2 = ₹13$
Similarly, we can calculate the cost of the pens for each Day and Carton combination. The table below presents the number of pens, along with the carton the pen is from, the price, cost and profits of the pens.

Day	Number of Pens	Revenue	Cost	Profit
Day 1	54 (C1)	1349	1109	240
	88 (C2)			
Day 2	34 (C2)	1271	879	392
	88 (C3)			
	2 (C4)			
Day 3	72 (C4)	999	747	252
	36 (C5)			
Day 4	84 (C5)	969	831	138
	30 (C6)			
Day 5	66 (C6)	945	742.5	202.5
	42 (C7)			
Day 6	22 (C7)	1040	878.5	161.5
	82 (C8)			

11. Profit made by Raghu on Day 4 = 63 + 75 = ₹138 Ans: (138)
12. On Day 1, the first 54 pens have profit percentage of more than 20%. All the pens sold on Day 2 have more than 20% profit percentage. On Day 3, the first 72 pens have profit percentage more than 20%. On Day 4, the 30 pens sold have a profit percentage more than 20%. On Day 5, the first 66 pens, and on Day 6, the first 22 pens have profit percentage more than 20%.
Therefore, the total number of pens = 54 + 34 + 88 + 2 + 72 + 30 + 66 + 22 = 368 Ans: (368)
13. Raghu made the lowest profit on Day 4. Choice (D)
14. The average profit made by Raghu per day from Day 3 to Day 6 = $\frac{252+138+202.5+161.5}{4} = 188.5$ Choice (C)
15. Profit was made by selling the pens in Carton 6 = 30 × 8.5 + 66 × 8.75 – 576 = ₹256.50. Choice (A)

Solutions for questions 16 to 20:

16. Given that n = 6.
First five numbers announced are 3, 1, 1, 0 and 1.
Since the first number announced is 3, the first ball cannot be Red or Yellow (since three balls must have been drawn in such a case).
Hence, the first ball can be Blue or Orange.
Let the first ball be Blue.
Since the number 3 was announced, there must be 3 yellow balls in the bag.
The second number is 1. The second ball cannot be Red or Yellow since no Orange or Red balls are drawn till then. The second ball cannot be Blue because there will still be 3 Yellow balls in the bag and if a Blue ball is drawn, the number announced must have been 3.

Hence, the second ball must be Orange.
If the second ball is Orange and this child announced 1, then there must be one Blue ball in the bag.
Hence, there will be only 3 yellow balls and 1 blue ball in the bag (since there must be a total of 6 balls).
The third child announced 1. The third ball can only be Yellow or Blue. In either case, the child cannot announce 1 (if he drew Yellow ball, he will announce 0; if he drew Blue ball, he will announce 3). Hence, this case is not possible.
Let the first ball be Orange.
Since this child announced 3, there must be 3 Blue balls in the bag. Also, three children announced the number 1. Since there are 3 blue balls in the bag, at least one child who drew a blue ball must have announced 1. Hence, there must be at least 1 yellow ball in the bag. ∴ There must be at least 1 orange ball, 1 yellow ball and 3 blue balls in the bag.
There can be at most one other ball in the bag.
The second ball can be Red or Blue (it cannot be Yellow or Orange as no Red ball was drawn and more than 1 blue ball is in the bag).
If the second ball is Red, then the six balls in the bag are one red, one yellow, one orange, and three blue balls. The third ball can be Yellow or Blue. If the third ball is Yellow, then the next three balls will all be Blue. However, the fourth, fifth and sixth numbers announced will all be 0 in this case. However, it is given that the fifth number is 1. Hence, the third ball cannot be Yellow. If the third ball is Blue, then there must be one Yellow ball in the bag for this child to announce 1. The fourth child must announce 0 and can only pick Blue or Yellow balls (since there are two Blue balls and one Yellow ball remaining in the bag). But for any of these colours, the number announced cannot be 0. Hence, this case is also not possible and the second ball cannot be Red.
If the second ball is Blue, the third ball can be Red or Blue. If the third ball is Red, the fourth ball must be Yellow or Blue (as they are the only colours of balls in the bag). In either case, the fourth child cannot announce 0. Hence, the third ball cannot be Red.
If the third ball is Blue, the fourth ball must be Yellow. Only then will the fourth child announce 0.
The fifth child could not have drawn the Blue ball and announced 1 (as there can only be one Yellow ball in the bag).
Hence, the fifth ball must be Red or Orange. The last ball must be the remaining Blue ball.
Hence, the order in which the balls are drawn is Orange, Blue, Blue, Yellow, Orange/Red, Blue.
The fourth ball to be drawn is Yellow in colour. Choice (C)

17. From the previous solution, the sixth ball to be drawn is Blue in colour. Choice (A)
18. Since the first child announced 1, it has to be Orange or Blue.
If it is Orange, there must be exactly one Blue ball in the bag. This case is not possible as only one Blue ball is drawn.
If the first ball is Blue, there must be one Yellow ball in the bag. For the child who draws the Yellow ball to announce 1, one Red ball must have been drawn. For the child who draws the Red ball to announce 1, one Orange ball must have been drawn. For the person who draws an Orange ball to announce 1, there must be one Blue ball remaining in the bag. Hence, these four balls must be drawn in the order Blue, Orange, Blue/Red, Red/Blue, Yellow. After the Yellow ball, four more Red balls can be drawn. In this case, a total of 9 balls will be drawn.
Hence, the maximum possible value of n is 9. Ans: (9)
19. The second ball is Orange. Choice (A)
20. The first child drew a Red ball. Since no Orange ball was drawn till then, he will announce 0.
The second child drew an Orange ball. Since there is one Blue ball in the bag, he will announce 1.
The third child drew an Orange ball. Since there is still one Blue ball in the bag, he will announce 1.
The fourth child drew a Blue ball. Since there are two Yellow balls in the bag, he will announce 2.
The fifth child drew a Yellow ball. Since one Red ball was drawn, he will announce 1.
The sixth child drew a Yellow ball. Since one Red ball was drawn, he will announce 1.
Therefore, the six children will announce the numbers 0, 1, 1, 2, 1 and 1. Choice (C)

Difficulty level wise summary - Section II	
Level of Difficulty	Questions
Very Easy	–
Easy	–
Medium	6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 20
Difficult	1, 2, 3, 4, 5, 16, 17
Very Difficult	18, 19

SECTION – III

Solutions for questions 1 to 22:

1. We can first consider a slightly simpler problem in which we remove the restriction that the two smaller groups should have an even number of coins. Also instead of 48 coins, we can consider, 2, 3, 4 or n coins.
Let n = 2. There is only one way to split it up, i.e., 1, 1. The product of these is 1, i.e., the sum of all the products, S = 1.
Let n = 3 this can be split up as 1, 2 and then 1, 1, 1
S = 1(2) + 1 = 3.
Let n = 4. The possible splits and the product are tabulated below.

Split	Product	Split	Product
1, 3	3	2, 2	4
1, 1, 2	2	1, 1, 2	1
1, 1, 1, 1	1	1, 1, 1, 1	

We see that irrespective of the way we split up, S = 6.
We can guess that for n, we get the (n – 1)[th] triangular number i.e., $\frac{n(n-1)}{2}$
(The numbers 1, 3, 6, 10... $\frac{n(n+1)}{2}$ are also called triangular numbers. These numbers represent the number of dots required to make grids in the shape of equilateral triangles of side represented by n dots).
Now we can consider the given problem where we start with an even number and at each stage split up the number into 2 even numbers.
Let n = 4, s = 4 = 4(1)
Let n = 6

Split	Product
2, 4	8
2, 2, 2	4

∴ s = 12 = 4(3)
Let n = 8

Split	Product
4, 4	16
2, 2, 4	4
2, 2, 2, 2	4

∴ s = 24 = 4(6)
If n = 48 = 2(24), S would be 4 times the 23[rd] triangular number S = $4\,\frac{23(24)}{2} = 1104$. Choice (B)

2. Let the ages of the five workers be a, b, c, d and e in the increasing order.
d + e = 116 → (1)
c + e = 114 → (2)
a + b = 102 → (3)
a + c = 105 → (4)
By adding the equations (1) and (3)
a + b + d + e = 218 → (5)
By adding all the 10 ages given in the problems (as each person's age occurs in four of the sums),
4(a + b + c + d + e) = 1092
a + b + c + d + e = 273 → (6)
subtracting (5) from (6), c = 55
putting c in (2), e = 114 – c = 59
Alternative solution:
As we have five equations and five unknowns, the problem is solvable. Hence Choice (D) is eliminated.
If e = 58, then d = 58. As the men have distinct ages, Choice (B) is eliminated.
If e = 60, d = 116 – 60 = 56
Now c = 54 and c + e = 110
But 110 is not present in the given values.
Hence, 59 is the only possible choice. Choice (C)

3. Each vessel is filled to 3/4 of its capacity. ∴ It is convenient to take the capacity as 4 units. Each vessel would have 3 units. In the second step, one-fourth the contents of W are transferred to H i.e., 1/4 units of milk and 3/4 units of water.
∴ It is convenient to take the capacity of each container as 16 units. The quantities at different stages are tabulated below.

	M			W			H		
	m	w	h	m	w	h	m	w	h
Initial	12				12				12
After step 1	8			4	12				12
After step 2				3	9		1	3	12
After step 3	8.25	2.25	3				0.75	2.25	9

Finally,
$\therefore m = \frac{8.25}{16}, w = \frac{9}{16}, h = \frac{9}{16}$ or m < w = h
Alternative solution:
It can be observed that the quantity milk leaving M is exactly 250 ml, while the quantity of water leaving W and that of honey leaving H is 750 × 1/4 = 187 ml each. Even though some amount of milk eventually returns to M (from H), it is clearly going to be very small to change the result in any significant way. Hence, M will have the least proportion of its initial content W and H will have an equal proportion. Choice (D)

4. If either a or b is divisible by p, it means that ab is divisible by p. Since ab + c is divisible by p, this implies c is divisible by p. Since c is a prime number, this is not possible. Hence, neither a nor b is divisible by p. Hence statement II is true. The remainder when ab is divided by p will be p – c but not c.
∴ Statement III need not be true.
Statement IV: $Rem\left(\frac{ab}{p}\right) + Rem\left(\frac{c}{p}\right) = p$
As $Rem\left(\frac{ab+c}{p}\right) = 0$, $Rem\left(\frac{ab}{p}\right) + Rem\left(\frac{c}{p}\right)$
= 0 or p or 2p
But 0 is not possible
For $Rem\left(\frac{ab}{p}\right) + Rem\left(\frac{c}{p}\right)$ to be equal to 2p,
$Rem\left(\frac{ab}{p}\right)$ or $Rem\left(\frac{c}{p}\right)$ must be greater than or equal to p, which is not possible.
∴ Only p is possible
∴ Statement IV is also true.
Hence, statements II and IV are true. Choice (C)

5. Let his cost price be ₹X and selling price be ₹Y.
If he had purchased for 12% less, then cost price would have been 88% of X and if he had then sold at 24% more, his selling price would have been 124% of Y.
Given that
$\frac{124\%Y - 88\%X}{88\%X} = \frac{5}{2}\left[\frac{Y-X}{X}\right]$
$\Rightarrow \frac{124Y}{88X} - 1 = \frac{5Y}{2X} - \frac{5}{2} \Rightarrow \frac{5Y}{2X} - \frac{62Y}{44X} = \frac{3}{2}$
$\Rightarrow \frac{48Y}{44X} = \frac{3}{2} \Rightarrow 16Y = 22X \Rightarrow \frac{Y}{X} = \frac{11}{8}$
∴ His actual profit percentage is $\frac{3}{8}(100\%)$
= 37.5% Choice (D)

6. Let the number of marbles with Amar at the end of the process be 3a. Now, the total number of marbles with Amar and Chetan = 40 – 10 = 30.
We tabulate the number of marbles with Amar, Bhavan and Chetan at different stages. We fill up the table from the bottom and see immediately that, initially the number of marbles with B is 10.

	A	B	C
Initial		10	
After A triples	a	30	10 – a
After B triples	3a(say)	10	30 – 3a

∴ Initial marbles with Bhavan = 10. Ans: (10)

7. Each man, each woman and each boy works for a total of 8 × 8, i.e., 64 days. Let Mi, Wi and Bi denote the work done by the i[th] man, the i[th] woman and i[th] boy. Let the total work be 1 unit.
$\sum_{i=1}^{8} 64\left[\frac{1}{M_i} + \frac{1}{W_i} + \frac{1}{B_i}\right] = 1$
$\sum_{i=1}^{8}\left[\frac{1}{M_i} + \frac{1}{W_i} + \frac{1}{B_i}\right] = \frac{1}{64}$
Hence the fraction of the work completed when the entire group works together for one day = 1/64 Choice (B)

8. 3x + xy = 138 – 2y
⇒ 2y + 3x + xy + 6 = 138 + 6
⇒ (y + 3) (x + 2) = 144, where x and y are integers greater than 0.
We have the following possibilities for x and y.

x	1	2	4	6	7	10	14	16	22	34
y	45	33	21	15	13	9	6	5	3	1

10 possibilities in all, as 144 can be written as product of 2 numbers in the following ways

(y + 3) × (x + 2)	(y + 3) × (x + 2)
1 × 144	16 × 9
2 × 72	18 × 8
3 × 48	24 × 6
4 × 36	36 × 4
6 × 24	48 × 3
8 × 18	72 × 2
9 × 16	144 × 1
12 × 12	

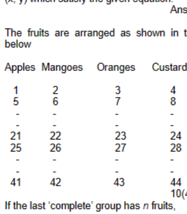
Since x > 0 and y > 0, y + 3 > 3 and x + 2 > 2
Thus we have 10 positive integral ordered pairs (x, y) which satisfy the given equation. Ans: (10)

9. The fruits are arranged as shown in the table below

Apples	Mangoes	Oranges	Custard-Apples
1	2	3	4
5	6	7	8
-	-	-	-
-	-	-	-
21	22	23	24
25	26	27	28
-	-	-	-
41	42	43	44
			10(45)

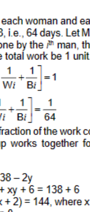
If the last 'complete' group has n fruits,
$\frac{n(n+1)}{2} \le 1000 < \frac{(n+1)(n+2)}{2}$
$\frac{(44)(45)}{2} = 990$ and $\frac{(45)(46)}{2} = 1035 \Rightarrow n = 44$
The mangoes occur in separate groups containing the following numbers 2, 6, 10, 14, 18, 22, 26, 30, ...
The sum of the first 7 of these (2 to 26) is 7[(2 + 26)/2] or 98. The 100[th] mango is the 2[nd] one in the 2[nd] group of in the 8[th] set (row).
The number of all the fruits upto that is [29(30)/2] + 2 or 437 Ans: (437)

10. Let us take a pentagon for example.



The area of pentagon is nothing but sum of the areas of the five triangles OAB, OBC, OCD, OCDE and OEA
Area of triangle = 1/2 × base × height
For all the triangles mentioned above height is radius of the circle 'r'.
∴ Area = 1/2 AB × r + 1/2 BC × r + 1/2 CD × r + 1/2 DE × r + 1/2 AE × r
$= r\left(\frac{AB + BC + CD + DE + EA}{2}\right) = rp$, where p is the half the perimeter of the polygon
It is given that the polygon is 100 sided and the lengths are 1,2.............100
$\therefore p = \frac{1 + 2 + ... + 100}{2}$
$= \frac{100 \times 101}{4} = 25 \times 101 = 2525$
∴ Area of polygon = rp = 2525r Choice (B)

11. Let (A, B), (C, D) and (E, F) be the pairs of wives and husbands. As we need a circular arrangement, we first fix the places of one among the six persons.



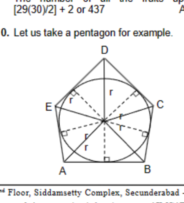
As there are five places to be taken by 5 people, the total number of ways in which 6 people can sit around a circular table = 5! = 120 ways.
Now, let us subtract the cases in which at least one pair of wife and husband are opposite each other
(i) The number of ways in which exactly one pair are opposite each other:
If B is opposite A, then C can take any one of the remaining 4 places and D can take any of the 2 remaining places (As D is not allowed to sit opposite C) and C and F can sit in the remaining two places in any order.
Thus, the number of ways = 4 × 2 × 2 = 16 ways.
Similarly, there will be 16 ways in which C and D are opposite each other and 16 ways in which E and F are opposite each other.
Thus, the total number of ways in which exactly one pair are opposite each other are 16 + 16 + 16 = 48 ways.
(ii) The number of ways in which exactly two pairs are opposite each other
As there are three pairs, if two pairs are opposite each other, then the third pair will also be opposite each other.
Thus it will be zero.
(iii) The number of ways in which exactly three pairs are opposite each other:
If B is opposite A, then C, can take any of the four places and D can take only one place, i.e., the place opposite C. E and F can occupy the remaining two places in any order.
Thus the total number of ways = 4 × 2 = 8 ways.
The number of ways in which no pair of wife and husband are opposite to each other = 120 – (48 + 8) = 64 ways. Ans: (64)

12. $a = 6^{1/2}, b = 6, c = 6^{3/2}, d = 6^2, e = 6^{5/2}$
$(a \times b \times c \times d \times e)^{20} = \left(6^{\frac{1}{2}+1+\frac{3}{2}+2+\frac{5}{2}}\right)^{20} = 6^{150}$
$\log 6^{150} = 150 \log_6 = 150(\log 2 + \log 3) = 150(0.7781)$
= 116.715
So the number of digits is 116 + 1 = 117. Choice (B)

13. Given N = DX + 23
And N = 12DY + 104
⇒ DX + 23 = 12D Y + 104
⇒ D(X – 12Y) = 81
∴ D must be a factor of 81, i.e., 1, 3, 9, 27 or 81
Since N leaves remainder 23 when divided by D, D > 23
∴ D = 27 or 81 ⇒ 6D = 162 or 486
Since, in both the cases 6D is greater than 104 and 6D is a factor of 120, the remainder when N is divided by 6D will be 104 itself. Choice (A)

14. Consider the following set of values of a, b, c, p, q, r.
a = 4, b = 16, c = 8
p = 3, q = 2, r = 1
$a^{q-r}\, b^{r-p}\, c^{p-q} = 4^{(2-1)}\, 16^{(1-3)}\, 8^{(3-2)}$
$= 4^1(16^1)\,(8^{-2}) = 1$
∴ None of choices A, B, C is true. Choice (D)

15.



ABCD is a cyclic quadrilateral
$= \frac{1}{2}(\angle BAD + \angle BCD) = \frac{1}{2}(180°) = 90°$ (i.e., angle in a semicircle)
⇒ ∠DAX + ∠DCY = 90° – (1)
⇒ But ∠DAY = ∠DCY (Angles made by same arc DY on the rest of the circle) – (2)
From (1) and (2)
∠DAX + ∠DAY = 90° ⇒ ∠XAY = 90°
⇒ XY is the diameter, and XY = 12 cm (given)
⇒ radius = 6 cm
∴ Area of the circle = π(6)²
= 36π sq.cm. Choice (A)

16. Given that $f(x + y) = \frac{f(x) + f(y)}{1 - [f(x)][f(y)]}$
$f(2n) = \frac{f(n) + f(n)}{1 - [f(n)]^2}$
Let f(n) = P ⇒ 1 – P² = 2P ⇒ P² + 2P – 1 = 0
$P = \frac{-2 \pm \sqrt{8}}{2} = -1 \pm \sqrt{2}$
Since P > 0, P = $\sqrt{2} - 1$
$f(3n) = \frac{f(2n) + f(n)}{1 - f(2n)f(n)} = \frac{1 + \sqrt{2} - 1}{1 - (1)(\sqrt{2} - 1)}$
$= \frac{\sqrt{2}}{2 - \sqrt{2}} = \frac{1}{\sqrt{2} - 1} = \frac{\sqrt{2} + 1}{2 - 1} = (\sqrt{2} + 1)$ Choice (D)

17. The minimum value will be realised when x = y = z = 1/3 and will be (1/3 + 3)² (3)
= (10/3)² (3) = 100/3 = 33 1/3.
If x, y, z are not taken as equal, say x = 0.3, y = 0.3, z = 0.4 the value of the expression is
$2\left(\frac{3}{10} + \frac{10}{3}\right)^2 + \left(\frac{2}{5} + \frac{5}{2}\right)^2 \approx 34.5$
∴ The minimum value is 33 1/3. Choice (C)

18. The circle which touches both the axes will be of the form (x – r)² + (y – r)² = r² (i.e., centered at (r, r))
Since, the given circle also passes through (3, 6),
(3 – r)² + (6 – r)² = r²
⇒ 9 + r² – 6r + 36 + r² – 12r = r²
⇒ r² – 18r + 45 = 0
∴ r = 3 or 15
∴ There exist two circles which satisfy the given conditions. Choice (C)

19. The distance covered by A and B and the time that they take to cover it are tabulated below.

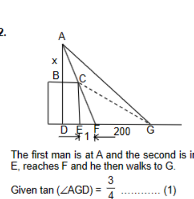
	A	B	Total
Distance	d_1	d_2	140 m
Time	t_1	t_2	20 sec

The average speed is 7 m/s. If it equals the average of the two speeds, either, the two speeds are equal (and equal to 7 m/s) or, the two time periods are equal. i.e. Either $\frac{d_1}{t_1} = \frac{d_2}{t_2} = 7$ m/s or t_1 = t_2 = 10 seconds
We consider each of the choices:
I. If A's speed is 6 m/s. B's speed has to be 8 m/s (so that the average can be 7 m/s) – True
II. If B's speed is more than 8 m/s (i.e., not 7 m/s) $t_1 = t_2$ = 10 sec, i.e., A runs for 10 sec which is less than 12 seconds. – True
III. If B's speed is 7 m/s, A's speed has to be 7 m/s – True
IV. If A's speed is less than 6 m/s (i.e. is not 7 m/s), $t_1 = t_2$ = 10 sec, i.e., B runs for 10 sec (and not more than 12 sec) – False
Hence, only IV is false. Choice (B)

20. We consider the following possibilities.
(a) 6 boxes are filled with a yellow ball. In this case, each of 3 boxes must be filled with a white or a black ball. These boxes can be chosen in 7 ways and can be filled in 2³ ways i.e., 8 ways for each chosen way. (∵ Each of the 3 boxes can be filled either with a white ball or a black ball)
∴ Total number of ways = 7(8) = 56
(b) 7 boxes are filled with a yellow ball.
In this case, each of 2 boxes must be filled with a white or a black ball. These boxes can be chosen in 8 ways and can be filled in 2 × 2 ways i.e., 4 ways for each chosen way.
∴ Total number of ways = (8) (4) = 32
(c) 8 boxes are filled with a yellow ball.
In this case, 1 box must be filled with a white ball or a black ball. This box can be chosen in 9 ways and can be filled in 2 ways.
∴ Total number of ways = 18
(d) 9 boxes must be filled with a yellow ball.
In this case, there is only 1 way of filling possible.
Total number of ways of filling = 56 + 32 + 18 + 1 = 107 Ans: (107)

21. Given DE is parallel to BC
⇒ Δ ADE is similar to ΔABC
Also given that AD = $\frac{1}{4}$(AB)
⇒ Area [ΔABC] = 16 × Area [ΔADE] = 16 × 5 cm² = 80 cm²
⇒ Area [trapezium BCED] = 80 – 5 or 75 cm²
In the trapezium BCED,
Area [ΔBFC] = 16 × Area [ΔDEF] [∵ BC = 4(DE)]
Let the area of ΔDEF be x.
⇒ Area [ΔBFC] = 16x
Area [ΔBDF] = Area [ΔCFE]
= $\sqrt{Area[\Delta DEF] \times Area[\Delta BFC]}$
= $\sqrt{x \times 16x}$ = 4x
∴ Area of the trapezium BCED
= x + 16x + 4x + 4x
= 25x = 75 cm² ⇒ x = 3 sq.cm
∴ Area [ΔCFE] = 4x = 12 sq.cm Ans: (12)

22.



The first man is at A and the second is initially at E, reaches F and he then walks to G.
Given tan (∠AGD) = $\frac{3}{4}$ .... (1)
and tan (∠CGE) = $\frac{1}{2}$ .... (2)
(2) ⇒ CE = 150 (∵ CE/EG = CE/300 = ½)
Δ ABC ~ ΔCEF
$\frac{x}{BC} = \frac{CE}{EF} = \frac{150}{100} = \frac{3}{2}$
∴ Let BC = 2h and x = 3h
(1) ⇒ $\frac{x + CE}{DE + 300} = \frac{3}{4}$
⇒ $\frac{3h + 150}{2h + 300} = \frac{3}{4}$ ⇒ h = 50
∴ x = 150 Ans: (150)

Difficulty level wise summary - Section III	
Level of Difficulty	Questions
Very Easy	–
Easy	6, 14
Medium	3, 5, 9, 12, 15, 17, 19, 21
Difficult	2, 4, 7, 10, 11, 13, 16, 18, 20, 22
Very Difficult	1, 8

Done
Tip: Scroll slowly for the best results